

STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

CANDIDATE ROLL NO: 2026-CS-PH-015
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BARCODE SCAN TRACKING



BK-PH-80015

Q.No	Q1	Q2	Q3	Q4	Q5
Max	14	14	14	14	14
Marks					

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

Q.1 Ans I. Group Velocity & Particle Velocity:

- A moving particle is represented by a de Broglie wave packet.

- Phase Velocity (v_p): Speed of a single wave = ω / k .

Since $\omega = E/\hbar$ and $k = p/\hbar$, $v_p = E/p$.

For non-relativistic particle, $E = p^2 / 2m \Rightarrow v_p = p / 2m = v/2$.

- Group Velocity (v_g): Speed of the wave packet envelope:

$v_g = d(\omega) / dk = dE / dp$.

$dE/dp = d(p^2 / 2m)/dp = p/m = v$ (particle velocity).

- Conclusion: Group velocity equals the particle velocity ($v_g = v$).

The wave packet spreads over time due to dispersion, which is described by Schrodinger's equation.

Q.2 Ans 2. Optical Interference & Alternative Strategies:

1. Basic Principle: Superposition of coherent light beams producing bright and dark fringes.

2. Two Main Methods:

- Division of Wavefront: Incident wavefront is split spatially using two apertures.

Example: Young's Double Slit (YDSE). Fringe separation: $\beta = \lambda \cdot D / d$.

- Division of Amplitude: Splitting amplitude via partial reflection and transmission.

Example: Newton's Rings and Thin Films.

3. Thin Film Interference:

Optical path difference = $2 \cdot \mu \cdot t \cdot \cos(r) \pm \lambda/2$.

Condition for constructive interference (Bright fringe): $2 \cdot \mu \cdot t \cdot \cos(r) = (2n - 1) \cdot \lambda/2$.

Condition for destructive interference (Dark fringe): $2 \cdot \mu \cdot t \cdot \cos(r) = n \cdot \lambda$.

4. Newton's Rings: Fringes of equal thickness formed between plano-convex lens and glass plate.

Diameter of nth dark ring: $D_n = \sqrt{4 \cdot n \cdot R \cdot \lambda}$.

Q.3 Ans 3. Lasers & Optical Fiber Technology:

A. Laser Fundamentals:

- Three processes: Absorption, Spontaneous Emission, Stimulated Emission.
- Einstein's relation: $B_{12} = B_{21}$ and $A_{21}/B_{21} = 8\pi h \nu^3 / c^3$.
- Population inversion: $N_2 > N_1$ achieved using an optical pumping source and metastable state.
- Laser characteristics: High coherence, monochromaticity, directionality, and high brightness.

B. Optical Fibres:

- Structure: Cylindrical dielectric waveguide with Core (n_1) and Cladding (n_2), $n_1 > n_2$.
- Total Internal Reflection guides light along the core.
- Numerical Aperture: $NA = \sqrt{n_1^2 - n_2^2} = \sin(\theta_{\text{max}})$.
- Applications: Telecommunication backbones, medical endoscopes, optical sensing.

Q.4

Ans 4. Electromagnetism and Maxwell's Field Equations:

1. Maxwell's Equations in Differential Form:

- 1. $\text{div}(\mathbf{D}) = \rho$ (Gauss Law for Electric field)
- 2. $\text{div}(\mathbf{B}) = 0$ (Gauss Law for Magnetic field)
- 3. $\text{curl}(\mathbf{E}) = - \frac{d\mathbf{B}}{dt}$ (Faraday's Induction Law)
- 4. $\text{curl}(\mathbf{H}) = \mathbf{J} + \frac{d\mathbf{D}}{dt}$ (Ampere's Law modified with Displacement Current $\mathbf{J}_d$)

2. Displacement Current: $\mathbf{J}_d = \epsilon_0 \frac{d\mathbf{E}}{dt}$. Maxwell added this term to maintain conservation of charge across capacitor charging cycles.

3. Boundary Conditions:

- $E_{1t} = E_{2t}$ (Tangential electric field continuous)
- $D_{1n} - D_{2n} = \rho_s$ (Normal displacement field)
- $B_{1n} = B_{2n}$ (Normal magnetic flux density continuous)
- $H_{1t} - H_{2t} = K$ (Tangential magnetic field)

4. Poynting Vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ gives the rate of energy transport by EM wave.

Q.5

Ans 5. Semiconductor Physics & Crystal Structure:

1. Crystal Lattices:

- SC (APF = 0.52), BCC (APF = 0.68), FCC (APF = 0.74).
- Miller Indices (hkl): Interplanar spacing $d_{hkl} = a / \sqrt{h^2 + k^2 + l^2}$.
- Bragg's Law: $2 * d * \sin(\theta) = n * \lambda$.

2. Band Theory:

- Periodic potential creates Valence Band, Conduction Band, and Energy Gap E_g .

3. Carrier Concentration:

- Total current density: $J = q * (n * \mu_e + p * \mu_h) * E$.
- Fermi-Dirac distribution: $f(E) = 1 / [1 + \exp((E - E_f)/kT)]$.

4. Direct vs Indirect Bandgap:

- Direct bandgap semiconductors (GaAs, InP) emit light efficiently upon electron-hole recombination.
- Indirect bandgap semiconductors (Si, Ge) involve lattice vibrations (phonons).

--- End of Answer Booklet (All 5 questions attempted) ---

Candidate Sign: Devansh